

MANUEL BOLAÑOS VALENCIA

Full-Stack Developer · AI Engineering · Embedded Systems & IoT

Castellón de la Plana, Spain | manuelb.disenio@gmail.com | +34 675 989 629

PROFESSIONAL SUMMARY

Full-stack web developer with over a decade of experience and 200+ delivered projects, with a growing side interest in embedded systems and IoT. My core is the web stack — PHP, Laravel, WordPress (professional), React, and JavaScript — with deep experience in CRM-driven enterprise development: custom solutions on SugarCRM, custom-endpoint integrations connecting the CRM to e-commerce platforms and third-party systems, and marketing automation on ActiveCampaign, including tailored integrations with other CRMs. I've delivered large-scale billing, ticketing, scheduling, and queue-management systems for clients in the automotive, security, and healthcare sectors, and I currently work on WordPress platforms for U.S. law firms in the legal marketing space. Along the way I've led teams as Scrum Master and Technical Lead under agile methodologies, and I actively integrate AI agents and LLMs into real products. Equally comfortable architecting a Laravel application, shipping a professional website, or wiring an LLM into an existing workflow.

WORK EXPERIENCE

Content Pilot — Dallas, Texas, USA

December 2022 – Present

Full Stack Developer

- Build custom WordPress and headless websites end to end — including bespoke themes and plugins tailored to client specific requirements — owning front-end development with React and JS frameworks alongside back end and database design.
- Develop custom property listing experiences with search, filtering, and detail views, plus end to end viewing scheduling and booking flows that keep availability, appointments, and confirmations in sync with third party reservation systems.
- Design custom API integration layers on top of WordPress — REST endpoints, GraphQL queries, response normalization, and caching — connecting external platforms to both traditional and headless front ends while keeping listing heavy pages fast on large datasets.
- Conduct website performance audits and optimizations, improving load times on content-heavy, image and data rich sites.
- Collaborate in Scrum ceremonies with developers, tech leads, and project managers to scope and deliver features.
- Provide ongoing support and troubleshooting for live websites, documenting solutions and tracking ticket resolution.

Stack: WordPress, Genesis Framework, PHP, JavaScript, React, HTML, CSS, Sass, MySQL, Webpack, SEO, Scrum, API Rest, GraphQL, XML, Schema, SEO, Cookies, Algolia, LLM

Existaya.com — Cali, Colombia

November 2020 – December 2022

Technical & Operations Leader

- Led IT operations and a team of developers, acting as Technical Lead and Scrum Master: defined processes, work schedules, and performance goals while coordinating delivery across web and UX teams under agile methodologies.
- Owned the company's CRM practice, delivering custom development on SugarCRM — modules, workflows, and business logic — and designing integrations with third-party platforms through custom REST endpoints that connected the CRM to e-commerce stores and internal business systems.
- Directed marketing automation projects on ActiveCampaign, including custom integrations between the platform and client CRMs to keep contacts, events, and sales pipelines in sync.
- Delivered enterprise software for clients in the automotive, security, and healthcare sectors: large-scale billing platforms, ticketing systems, and appointment/queue-management systems.
- Reported progress and metrics to the COO and key stakeholders, managed vendor and client relationships, and contributed to business expansion initiatives (partnerships, acquisitions, and corporate alliances).

Stack: WordPress, PHP, JavaScript, Angular, React, October CMS, Shopify, VTEX, WooCommerce, Sass, MySQL, REST, CRM, Sugar CRM, ActiveCampaign, Elastic Search.

Existaya.com — Cali, Colombia

January 2016 – November 2020

Project Manager / Technical Leader

- Managed web projects and a 5-person development team from requirements through delivery.
- Coordinated Scrum ceremonies and ensured projects met scope, quality, and deadlines.
- Built and maintained solutions across PHP/WordPress, React, Angular, and e-commerce platforms (Shopify, VTEX, WooCommerce).
- Delivered progress reports and presentations to stakeholders and the COO.

Stack: WordPress, PHP, JavaScript, Angular, React, Vue, October CMS, Shopify, VTEX, WooCommerce, Sass, MySQL, REST

Existaya.com — Cali, Colombia

February 2012 – January 2016

Web Developer / UX Developer

- Designed and built responsive websites and user interfaces (HTML5, CSS3, JavaScript, jQuery, WordPress).
- Integrated multimedia content and conducted website performance and UX testing.
- Collaborated with designers, developers, and stakeholders to deliver client projects.

Stack: WordPress, PHP, JavaScript, HTML, CSS, Angular, React, PhoneGap, IONIC

EME — Cali, Colombia

January 2011 – December 2011

Web Developer

- Developed and maintained websites using HTML5, CSS3, JavaScript, PHP, and WordPress.
- Built user interfaces, integrated multimedia content, and tested site performance.

Stack: WordPress, PHP, JavaScript, HTML, CSS

PROJECTS — EMBEDDED SYSTEMS, IOT & AI

4WD Exploration Rover (ESP32) — personal build

- Designing and building a four-wheel-drive exploration rover around ESP32 microcontrollers, with a custom suspension system and a self-designed hexagonal PETG chassis.
- Implementing skid-steer motor control via BTS7960 drivers, a phased modular firmware architecture, and FPV video streaming.
- Working across mechanics, electronics, and embedded firmware (C/C++) within a single project.

Embedded & IoT Experiments (ESP32) — personal build

- Built WiFi-connected applications on ESP32 boards, including a live news ticker (NewsAPI + ArduinoJson) with category rotation on a TFT display.
- Interfaced TFT LCDs (ST7735) over SPI using the TFT_eSPI library; hands-on sensor integration and display driver troubleshooting.

AI-Assisted Development & LLM Integration

- Integrating LLMs and agent-style workflows into applications.
- Using AI-assisted tooling (Claude Code with MCP servers — GitHub, MySQL, filesystem, Playwright, Figma) to accelerate development.

TECHNICAL SKILLS

Backend: PHP, Laravel, WordPress, October CMS, MySQL, PostgreSQL, REST / SOAP / GraphQL APIs.

Front-end: JavaScript, React, Vue, Angular, HTML5, CSS3, Sass (BEM), Bootstrap 5, Tailwind.

E-commerce: Shopify, VTEX, WooCommerce.

Embedded & IoT: ESP32, Arduino, C, C++, firmware development, SPI / I²C, sensors & displays, motor drivers (BTS7960).

AI: LLM integration, AI agents, prompt engineering, AI-assisted development (Claude Code, MCP).

Tooling & DevOps: Git / Git Flow, Webpack, Google Cloud (GCP), Netlify, Linux / macOS / Windows.

Methodologies: Scrum (Scrum Master / Product Owner), Agile, technical leadership, project & team management.

Languages: Spanish and English.

EDUCATION & CONTINUOUS LEARNING

ICESI University — Cali, Colombia · 2016

Diploma, Project Management (PMI)

CECEP Technology University — Cali, Colombia · 2012 – 2015

Degree, Systems Technologist / Software Developer

CECEP Technology University — Cali, Colombia · 2015

Diploma, Information Security / Project Management

FADP Institute — Cali, Colombia · 2008 – 2011

Degree, Graphic & Web Design

Online (Platzi & Udemy) · Ongoing

Laravel, WordPress, React, Angular, JavaScript (ES6), Git, MySQL, Shopify, VTEX

Continuous Learning — Embedded Systems & Robotics

Self-directed path: CS50x, C / C++, Arduino, ESP32 / IoT — progressing toward ROS 2 and STM32.